

Muhsina Tarannum Munfa

PhD Applicant | AI/ML Researcher,

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RESEARCH INTEREST

My research interests lie in low-resource and multilingual natural language processing, AI for healthcare, and explainable machine learning. I am particularly interested in developing robust, human-centered AI systems for underrepresented languages and real-world applications.

EDUCATION

Metropolitan University

BSc in Computer Science & Engineering

GPA: 3.56/4.00

2018 - 2022

Sylhet, Bangladesh

PUBLICATIONS

Md Sultanul Islam Ovi, **Muhsina Tarannum Munfa**, G.M.M Miftahul Alam Adib, Syed Sabbir Hasan (2026). **A Dual Pipeline Machine Learning Framework for Automated Multi Class Sleep Disorder Screening Using Hybrid Resampling and Ensemble Learning** (arxiv) 

Under review: *Social Network Analysis and Mining*

- Implemented and evaluated machine learning models for automated multi-class sleep disorder prediction.
- Developed data preprocessing, model training, and performance evaluation pipelines using Python and scikit-learn.
- Contributed to the study design and early manuscript development, including drafting the initial methodology.

RESEARCH EXPERIENCE

ResearchDen

Research Collaborator

2025 - Present

Remote

- Contributed to the initial research design and manuscript preparation for multi-class sleep disorder prediction
- Contributing to dataset creation, preprocessing, and experimental design for predictive modeling studies.
- Assisting in the development and evaluation of explainable AI (XAI) methods for healthcare applications.

Prediction of Customers' Purchase Intention in E-commerce Stores

B.Sc. Thesis | Metropolitan University | Supervisor: Akhlak Uz Zaman Ashik, Assistant Professor

Jan 2022 – Aug 2022

Sylhet, Bangladesh

- Investigated customer purchase intention prediction using machine learning on an e-commerce dataset
- Implemented and compared Logistic Regression, KNN, SVM, Naive Bayes, Decision Tree, Random Forest, and ANN models.
- Performed data preprocessing, feature engineering, and model evaluation, identifying Random Forest and ANN as the best-performing models.

WORK EXPERIENCE

FlyRank AI

Backend AI Engineering Intern

Jul 2026 – Present

Remote

- Collaborating with the engineering team on AI system implementation and backend development.
- Developing backend components for AI-powered applications using Python and modern API frameworks.

YES Associate

Admission Support Officer

2023 - 2024

Sylhet, Bangladesh

- Prepared and reviewed application documents for students applying to international universities.
- Reviewed student files for completeness and compliance with application requirements.
- Assisted applicants throughout the admissions documentation and application process.

OpenGenus IQ

Research Scientist, Intern

2022 - 2023

Remote

- Conducted literature reviews on computer science and machine learning topics using technical and educational resources.
- Produced technical articles by synthesizing information from multiple sources into clear explanations.
- Developed technical writing and research communication skills through independent topic exploration.

PROJECTS

Curriculum-Based Bangla QA System | Project Lead  | ongoing ChromaDB | FastAPI | LLM

- Leading the development of an NLP system for generating Bangladeshi curriculum-grounded exam questions
- Coordinating a team-based implementation using Python, ChromaDB, LLMs, and Bangla NLP models
- Designing the retrieval, question-generation, and answer-validation pipeline to improve factual grounding and curriculum alignment

Bangla Voice Assistant for Low-Literacy Users  | ongoing BanglaSpeech2Text | gTTS

- Selected BanglaSpeech2Text for server-side speech recognition
- Trained Lightweight classifier and mapped intent-to-document
- Applied gTTS for Text-to-Speech

Air Quality & Health Risk Assessment System | Live Demo  FastAPI | PostgreSQL | XGBoost | Streamlit | Looker Studio

- Built an end-to-end system for AQI prediction, integrating external APIs, XGBoost, PostgreSQL, and FastAPI
- Configured PostgreSQL on Neon with schema for time-series AQI storage and managed deployment on Render
- Developed interactive Streamlit and Looker Studio dashboards for real-time air quality monitoring and health risk visualization.

ChefBot | Live Demo  LangChain | ChromaDB | Sentence Transformers | Groq API | Gradio

- Developed a RAG-based cooking assistant using LangChain, ChromaDB, and LLaMA 3.
- Implemented document ingestion, embedding generation, and retrieval pipelines for cookbook-based question answering.
- Reduced hallucinations through prompt engineering and context-constrained response generation.

DATASET PROJECTS

AQI Dataset for Sylhet, Bangladesh 

- Derived from a larger environmental dataset
- Cleaned records, calculated AQI values, and labeled air quality categories to support analytics and predictive modeling
- Published the processed dataset on Kaggle with documentation for reproducible use.

Bangladesh Climate and Agriculture Dataset 

- Curated and standardized climate and agricultural data from multiple publicly available sources
- Cleaned, validated, and documented the dataset to support machine learning and data analysis tasks.
- Published the processed dataset on Kaggle with comprehensive metadata and usage documentation.

SKILLS

- **Programming Languages:** Python, C++, JavaScript
- **Machine Learning & Deep Learning:** Scikit-learn, HuggingFace Transformers PyTorch
- **NLP & LLM:** LangChain, ChromaDB, FAISS, Groq API
- **Backend & Data Engineering:** FastAPI, PostgreSQL, MongoDB, REST APIs, SQL
- **Deployment & MLOps:** Streamlit, Gradio, Hugging Face Spaces, Render
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn,
- **Tools & Platforms:** Git, GitHub, Google Colab, VS Code, Looker Studio, LaTeX, Typst

ACADEMIC SERVICES

Conference Reviewer | 2026

- NGEN-AI 2026, Italy
- QUANCOM 2026, Italy

Peer Reviewer | 2026

- European Journal of Computer Sciences and Informatics

CERTIFICATIONS

- **Research Project Lifecycle Workshop** | Organized by AI/ML Professional Community Bangladesh October - 2025
- **Research Scientist, Intern at OpenGenus** April -2023